

MSc Data Science Project

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Department of Physics, Astronomy and Mathematics

Data Science FINAL PROJECT REPORT

Project Title:

Exploring Renewable Energy and Energy Efficiency Impacts on
Sectoral Economic Growth in Low- and Middle-Income Countries.

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DECLARATION STATEMENT

This report is submitted in partial fulfilment of the requirement for the degree of Master of Science **in Data Science** at the University of Hertfordshire.

I have read the detailed guidance to students on academic integrity, misconduct and plagiarism information at [Assessment Offences and Academic Misconduct](#) and understand the University process of dealing with suspected cases of academic misconduct and the possible penalties, which could include failing the project or course.

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5.1 Tools, Libraries, and Environment

To conduct this comprehensive analysis, I worked extensively in Python, alternating between Jupyter Notebook for iterative local development and Google Colab for resource-intensive tasks such as rendering large heatmaps and geospatial plots. This hybrid setup allowed me to optimize both speed and flexibility depending on the computational needs of each task.

I employed a wide suite of powerful Python libraries, each carefully selected based on their capabilities in data science and machine learning:

- Pandas and NumPy served as my core tools for managing and transforming large-scale, multi-year, multi-indicator time-series data. I used them for reshaping, merging, missing value handling, and statistical computations.
- Matplotlib and Seaborn were used for crafting an extensive range of visualizations — including box plots, violin plots, multi-type time series trends, histograms, correlation heatmaps, PCA scree plots, and clustering diagrams. I relied on Seaborn's expressive color palettes (e.g., "coolwarm", "Set2") for visual clarity.
- Scikit-learn (sklearn) was the backbone of my machine learning workflow. I used it for data standardization (StandardScaler), dimensionality reduction (PCA), unsupervised learning (KMeans clustering), and internal validation metrics (silhouette score).
- For geospatial visualization, I turned to GeoPandas in combination with Natural Earth shapefiles (ne_110m_admin_0_countries.shp). This enabled me to map country clusters geographically and observe regional patterns that transcended income level boundaries.

Regarding data sources, I used two distinct datasets:

- The primary dataset was a World Bank-derived CSV file containing 91 economic, energy, and environmental indicators per country from 1989 to 2023. This dataset was pre-structured into columns such as "Series Name", "Country Code", and yearly values (e.g., "1990 [YR1990]", "2023 [YR2023]"), which I then cleaned and transformed for analysis.
- The income classification dataset was retrieved from the official World Bank classification portal via the URL:
<https://databank.worldbank.org/data/download/site-content/CLASS.xlsx>.
This Excel sheet contained yearly income categorizations such as "Low income", "Lower middle income", and "Upper middle income" for all recognized countries.

In total, this environment and toolchain enabled me to maintain a consistent, modular, and highly replicable analytical pipeline from preprocessing through to final clustering and visualization stages.

5.2 Data Preprocessing

To prepare the dataset for rigorous, income-level-based analysis, I undertook a series of highly structured preprocessing steps. I combined technical precision with logical checks to ensure the dataset was reliable, standardized, and robust for all downstream statistical and machine learning procedures.

I began by importing essential libraries — NumPy, Pandas, and Matplotlib — to facilitate numerical operations, dataframe manipulation, and initial inspections. I loaded the raw dataset from the World Bank repository using the `.read_csv()` function and immediately addressed the issue of placeholder values ("..") by replacing them with `pd.NA`, thereby marking them as true missing values in Pandas.

Next, I isolated only the year-specific columns starting from the 5th column onwards using `df.columns[4:]`, and applied `pd.to_numeric()` with `errors='coerce'` to convert all these values into float-compatible formats. This ensured consistent numerical interpretation of values across time, coercing any remaining anomalies into `NaN`.

To impute these missing values meaningfully, I applied group-wise forward and backward filling — first by grouping the dataframe by ["Country Name", "Series Name"] and then transforming the year columns using a chained `.bfill().ffill()` method. This technique maintained the integrity of time-series continuity, ensuring that missing values within a country's series were filled only using that same country-series context.

While working on post-fill quality, I noticed some anomalous values in the 2023 column. To avoid inflated values corrupting my trend analysis, I inserted a rule-based filter where any 2023 value greater than $1.5\times$ of its corresponding 2022 value was replaced with the 2022 value. This step was implemented using the `.where()` method:

```
df["2023 [YR2023]"] = df["2023 [YR2023]"].where(df["2023 [YR2023]"] < df["2022 [YR2022]"] * 1.5, df["2022 [YR2022]"])
```

Following this, any remaining `NaN` values were imputed using the row-wise mean of each country-series pair, implemented via `.apply(lambda row: row.fillna(row.mean()), axis=1)`. This strategy preserved the contextual distribution of each individual row rather than borrowing information across rows or columns.

To avoid distortions from sparsely populated series, I calculated a missing-value threshold set at 50% of the number of year columns. Any row (i.e., country-indicator pair) with missing values exceeding this threshold was dropped entirely using `df.dropna(thresh=missing_threshold, axis=0)`, ensuring that only meaningful rows were retained for downstream analysis.

Once my base dataset was cleaned, I moved to merge it with the World Bank's income classification dataset for 2024–2025, sourced from:

<https://databank.worldbank.org/data/download/site-content/CLASS.xlsx>.

I used `pd.read_excel()` with `skiprows=3` and custom column names to ingest the sheet titled "List of economies", extracting only ["Country Name", "Country Code", "Income Level"] for the merge. Using an inner join on both "Country Name" and "Country Code" allowed me to accurately map each country to its respective income group.

After the merge, I identified unmatched records — i.e., countries with NaN in "Income Level" — using logical filtering. Many of these were regions or aggregates, not actual countries, such as "World", "Euro area", or "Sub-Saharan Africa". I compiled a comprehensive list of these aggregate entries and removed them using `~df["Country Name"].isin(regions_to_remove)`. This filtered the dataset to country-level rows only.

Still, a handful of actual countries remained unmatched due to naming mismatches or edge-case classifications. I reviewed these manually and constructed a mapping dictionary of corrected "Income Level" values using domain knowledge and World Bank classification updates. For instance:

```
manual_income_levels = {
    "Cote d'Ivoire": "Lower middle income",
    "Sao Tome and Principe": "Lower middle income",
    "Turkiye": "Upper middle income",
    "Viet Nam": "Lower middle income",
    "Afghanistan": "Low income",
    "Venezuela, RB": "Not classified",
    ...
}
```

I applied this mapping using `.map()` and `.fillna()` to ensure a fully populated "Income Level" column. Next, I explicitly excluded high-income countries and "Not classified" entries using a final logical filter. This brought the dataset down to only three groups: Low income, Lower middle income, and Upper middle income.

To validate data quality, I conducted an assertive check for duplicates using `.duplicated().sum()` and confirmed that no duplicates remained. Additionally, I verified that the "Income Level" column contained only the three intended classifications via an assertion check:

```
assert all(income in valid_income_levels for income in unique_income_levels)
```

As a final step, I incorporated expert recommendation updates. For instance, I removed "Bulgaria", "Palau", and "Russian Federation" manually, as these countries had just transitioned to the high-income bracket (World Bank, 2025). This filtering ensured temporal consistency between our data and the latest World Bank classification.

I then saved the fully processed dataset to disk using `.to_csv()`, ensuring reproducibility. From this point forward, the resulting dataset was clean, structured, and enriched with

income group metadata — making it ready for exploratory, statistical, and machine learning-based analyses

5.3 Exploratory Data Analysis (EDA)

After preparing the dataset, I transitioned into exploratory analysis to derive meaningful insights. My EDA process was comprehensive and multi-layered, spanning descriptive statistics, visual exploration, time-series tracking, and correlation analysis. I ran the analysis using a combination of Pandas, Seaborn, Matplotlib, and NumPy, all within the Jupyter environment.

Thematic Correlation Analysis: Linking All Four Research Dimensions

To directly address my research question, I compiled four indicator blocks:

1. Renewable energy adoption
2. Energy efficiency
3. Sectoral economic growth
4. Emissions and carbon impact

I manually curated lists of relevant series under each block (e.g., *Renewable energy consumption (% of total final energy consumption)*, *Energy use per \$1,000 GDP*, *Services value added growth*, *Total greenhouse gas emissions excluding LULUCF*, etc.). These were combined into one master list: `all_research_series`.

Sector-Based Correlation Analysis

Lastly, I explored how CO₂ emissions, energy use, and renewables influenced sectoral growth in Agriculture, Industry, and Services. I defined a dictionary `sector_series_dict` and looped over each income level and sector.

For each combination, I:

- Filtered the dataset by income and selected sector-relevant indicators
- Averaged year-wise values to derive `Series Avg`
- Created a country-series pivot table using `pivot_table()`
- Computed three different correlation matrices — Pearson, Spearman, and Kendall — for each sector
- Visualized each matrix using `sns.heatmap()` with customized formatting

This yielded a total of 27 heatmaps (3 sectors × 3 income levels × 3 correlation types). Combinations with insufficient data (`pivot_df.shape[0] < 3`) were automatically skipped to preserve statistical integrity. The multi-method approach helped me triangulate sectoral insights and reduce the risk of over-reliance on a single correlation paradigm.

This exhaustive EDA served not only as the diagnostic phase of my project but also laid the groundwork for dimensionality reduction and cluster-based grouping in later stages. It helped me identify structural similarities across countries, changes over time, and thematic patterns across income classes.

5.4 Principal Component Analysis (PCA)

Once I had established thematic patterns from EDA, I transitioned into dimensionality reduction using Principal Component Analysis (PCA) to reveal latent structures and compress multivariate relationships across income levels. I handled PCA separately for Low, Lower-middle, and Upper-middle income countries, treating them as independent data universes.

To begin, I manually compiled a list of 17 indicators by merging four blocks—renewable energy, energy efficiency, sectoral economic growth, and CO₂ emissions—that mapped to the four core themes of my research question. I selected these series to capture a holistic yet compact representation of energy-growth-emission interdependencies.

For each income group:

- I filtered the master dataset to retain only countries from the target income level and only rows where Series Name belonged to selected_series.
- Next, I converted all year columns to numeric, dropping any parsing-related issues using errors='coerce', ensuring every series was analysis-ready.
- I computed a row-wise mean across all years to derive a time-agnostic column called "Series Avg" for every indicator-country pair, using .mean(axis=1).
- Using pivot_table(), I transformed the dataframe so that each row represented a country, and each column a research-relevant indicator. I excluded countries that had missing data across any of the selected series by applying .dropna().

Then, I applied standardization using StandardScaler() from scikit-learn to ensure each series was centered and scaled (mean=0, variance=1). This step was crucial to prevent high-magnitude indicators from dominating the PCA components.

I fit PCA on the scaled matrix with PCA().fit_transform() to reduce the dataset into a new orthogonal coordinate system. From this:

- I captured explained variance ratios of each principal component
- I generated a scree plot showing the cumulative variance explained, helping visualize how much information could be preserved using only two components

Finally, I constructed a two-dimensional PCA scatter plot, showing countries in the new PC1–PC2 space using sns.scatterplot(). I called plot_pca_with_labels() to annotate each point with its country name, enhancing interpretability.

After generating PCA plots, I aimed to go beyond visual clustering by quantifying inter-country similarity. For this, I computed a Euclidean distance matrix between countries using the original PCA input space (i.e., standardized Series Avg vectors). I implemented this through pdist() and converted it into a squareform matrix.

To enhance visual clarity and interactivity, I created an interactive Plotly heatmap using `px.imshow()`, which displayed country-wise distances inside each heatmap cell and allowed zooming and panning. These plots were saved in HTML format for documentation and easy sharing.

Through this, I uncovered patterns of intra-income group heterogeneity — for instance, while some Low-income countries clustered tightly (e.g., Chad and Congo Dem. Rep.), others like Ethiopia or Afghanistan appeared distant.

This helped me realize that not all countries in the same income level behave the same way, which challenges the assumptions behind income-level grouping and justifies my use of PCA and clustering.

5.5 Clustering and Profile Analysis

Building on the PCA foundation, I conducted unsupervised clustering using the KMeans algorithm on the PCA coordinates for each income group. I designed a function `run_kmeans_clustering()` that:

- Accepted PCA coordinates and maximum k (default 12)
- Iteratively ran KMeans from $k = 2$ to optimal, storing inertia and silhouette scores
- Chose optimal k using the maximum silhouette score, then fit a final KMeans model to label each country

I visualized the elbow curve to show inertia reduction per cluster count, helping support the chosen k . Then, I plotted the final cluster results in the PC1–PC2 space using cluster-wise scatter plots with text annotations for each country. These helped visually evaluate clustering separation and density.

To understand what defined each cluster, I profiled them using the original (non-PCA) indicators:

- I reloaded the original indicators per country using a filter on Income Level and countries used in PCA
- I grouped by cluster label and calculated cluster-wise mean values for all indicators
- I normalized those averages using min–max scaling, and generated radar-like line plots to contrast profiles across clusters visually

Next, I implemented geographic mapping of clusters using GeoPandas. I downloaded the Natural Earth shapefile for country boundaries and merged it with my cluster data. I corrected known naming mismatches between World Bank names and shapefile names using a manual `rename_map` dictionary. Then:

- I plotted each country shaded by cluster assignment
- I annotated maps with country names using centroid coordinates, keeping label sizes readable
- I repeated this for all income levels, generating three separate maps

Finally, to facilitate comparative interpretation, I wrote a composite function `plot_all_income_levels_geoclusters()` that:

- Mapped the clustered countries side-by-side using subplots(1, 3)
- Harmonized plotting scale, colouring, and legend handling
- Created a visual dashboard of global clustering outcomes by income level

This multifaceted clustering approach allowed me to:

- Identify latent country groupings beyond income level
- Profile each group in terms of indicator performance
- Visualize global geospatial patterns in energy-growth-emission profiles

Together, PCA and KMeans uncovered hidden structures that would have been lost in standard analysis, making them essential for the interpretative depth of this project.

5.6 Panel Regression Methodology

To rigorously examine the sectoral economic growth drivers across income groups, I employed panel data regression techniques using both Fixed Effects and Random Effects models. My goal was to quantify the influence of selected energy, emissions, and infrastructure indicators on the growth of agriculture, industry, and services sectors.

Library and Environment Setup

I used Python in a Jupyter environment, leveraging the `pandas` and `numpy` libraries for data manipulation, and `linearmodels` for estimating panel regressions. Additionally, I imported `statsmodels.api.add_constant` for including intercept terms, and `scipy.stats` for executing the Hausman test.

```
from linearmodels.panel import PanelOLS, RandomEffects
from statsmodels.tools.tools import add_constant
from scipy import stats
```

Selection of Target Variables

I focused on three key sectoral growth indicators from the World Bank:

- Agriculture, forestry, and fishing, value added (annual % growth)
- Industry (including construction), value added (annual % growth)
- Services, value added (annual % growth)

Each served as a dependent variable in a separate model. For each sector, I executed regressions individually for low-income, lower-middle-income, and upper-middle-income countries.

Independent Variable Selection

I selected the following five predictors based on domain knowledge, EDA, and correlation analysis:

1. Renewable energy consumption (% of total final energy consumption)
2. Energy intensity level of primary energy (MJ/\$2017 PPP GDP)
3. Access to electricity (% of population)
4. GDP per unit of energy use (PPP \$ per kg of oil equivalent)
5. Carbon dioxide (CO₂) emissions excluding LULUCF per capita (t CO₂e/capita)

These features were consistently used across all sectoral models. I later experimented with adding GDP growth (annual %) and PM_{2.5} air pollution based on their potential relevance and availability.

Data Preparation

I defined a reusable function `prepare_panel_data()` that extracted and reshaped the panel dataset for a given income level and dependent variable. This function:

- Filtered countries by income group
- Selected only relevant series names
- Melted the dataset from wide to long format
- Pivoted it to ensure each row represented a (country, year) pair
- Dropped missing values
- Converted year to integer

```
df_long = df_filtered.melt(id_vars=['Country Name', 'Series Name'],
var_name='Year', value_name='Value')
df_panel = df_long.pivot_table(index=['Country Name', 'Year'],
columns='Series Name', values='Value').reset_index()
```

I renamed columns for clarity and ensured the dataset had no nulls before regression.

Regression Execution

For each prepared dataset (e.g., `agri_df_low`), I used `PanelOLS` to fit the Fixed Effects model and `RandomEffects` to estimate the Random Effects model. Both models were fitted with robust standard errors to ensure valid inference under heteroskedasticity.

```
fe_model = PanelOLS(y, X,  
entity_effects=True).fit(cov_type='robust')  
re_model = RandomEffects(y, X).fit(cov_type='robust')
```

Hausman Test and Model Choice

To determine whether the Fixed or Random Effects model was more appropriate, I implemented the Hausman test manually. I computed the difference in coefficients and their covariance matrices to derive the test statistic and p-value.

```
b_diff = fe_results.params - re_results.params  
cov_diff = fe_results.cov - re_results.cov  
stat = b_diff.T @ np.linalg.inv(cov_diff) @ b_diff
```

If the p-value was below 0.05, I preferred the Fixed Effects model; otherwise, I used Random Effects. The model selection logic and summaries were printed for transparency.

Model Evaluation and Visualization

After model fitting, I extracted the predicted values and added them back to the original dataset for comparison. I then plotted the actual vs predicted growth for one representative country per income group, using `matplotlib`.

```
plt.plot(df_sample['year'], df_sample['actual'], label='Actual',  
marker='o')  
plt.plot(df_sample['year'], df_sample['predicted'],  
label='Predicted', marker='x')
```

PCA-based Validation

As a final methodological step, I conducted robustness validation by restricting the dataset to countries selected through prior Principal Component Analysis (PCA). These countries were identified as structurally representative within their income group based on energy-emission-growth profiles.

To implement this:

- I loaded PCA coordinate files for each income group.

- Then I filtered the main dataset to retain only those countries present in the PCA coordinate set.
- Using the filtered subset, I reconstructed the panel datasets and reran the same fixed and random effects models for a single sector and income group as a targeted diagnostic test, keeping all variable selections and model specifications unchanged.

6. Results

6.1 Correlation Heatmap for Sectoral Economic Growth

I examined the relationships between agriculture sector growth and environmental-energy indicators across low-income countries using Pearson, Spearman, and Kendall correlation methods. Across all approaches, there was a consistent negative correlation between agricultural value added and CO₂ emissions per capita (Pearson: -0.55, Spearman: -0.39, Kendall: -0.43), suggesting that higher emissions may be linked to reduced agricultural growth. Renewable electricity output also showed a strong negative association (Pearson: -0.56, Spearman: -0.61, Kendall: -0.33). In contrast, electricity production from renewable sources (excluding hydro) exhibited positive correlations, particularly under Spearman (0.48), indicating a potential supportive role of renewables. Combustible renewables and waste displayed a modest positive relationship across all methods.

For the industry sector in the same income group, industry growth was strongly and positively correlated with energy use per \$1,000 GDP (Pearson: 0.81), carbon intensity of GDP (0.42), and total greenhouse gas emissions (0.46), suggesting that energy-intensive and emission-heavy activities might be driving industrial output. The Spearman and Kendall results reinforced these associations, particularly in relation to energy use and carbon indicators. At the same time, GDP per unit of energy use had consistently negative correlations across Pearson (-0.54), Spearman (-0.46), and Kendall (-0.24), indicating lower energy efficiency in rapidly growing industries. The relationship between manufacturing growth and overall industry growth was also strong (Pearson: 0.74, Spearman: 0.96, Kendall: 0.90), reinforcing their interdependence.

In the services sector, energy use per \$1,000 GDP maintained a strong positive correlation with growth (Pearson: 0.82, Spearman: 0.68, Kendall: 0.52), while GDP per unit of energy use displayed a strong negative relationship (Pearson: -0.72, Spearman: -0.68, Kendall: -0.52). This indicated that higher energy intensity may be contributing to services expansion. Additionally, renewable energy consumption and renewable electricity output showed moderate positive correlations across all methods. On the other hand, fossil fuel energy consumption was negatively associated with services growth (Pearson: -0.63), particularly under Spearman (-0.64) and Kendall (-0.62), highlighting the influence of cleaner energy sources on sectoral development.

Shifting to lower middle-income countries, the correlations for the agriculture sector were generally weaker. Electricity production from renewable sources showed a consistent negative association with agriculture (Pearson: -0.29, Spearman: -0.23, Kendall: -0.17), indicating limited integration of renewables in this area. Other indicators mostly demonstrated weak or negligible correlations. Energy use per \$1,000 GDP and energy intensity had weak negative relationships, while GDP per unit of energy use showed a small

positive correlation (Pearson: 0.08, Kendall: 0.12). Greenhouse gas emissions, however, had slight positive correlations in Spearman (0.25) and Kendall (0.18), though not strongly influential.

In the industry sector, the results were mixed. CO₂ emissions excluding LULUCF consistently had a moderate negative correlation with industrial growth (Pearson: -0.46, Spearman: -0.44, Kendall: -0.31), suggesting that emissions may hinder expansion. Carbon intensity and fossil fuel energy use followed similar weak to moderate negative trends. Notably, combustible renewables and renewable energy consumption had positive correlations (Pearson: 0.38 and 0.35), pointing to a potentially supportive role for renewables. The connection between manufacturing and industry growth remained moderately strong across Pearson (0.62), Spearman (0.76), and Kendall (0.60).

As for the services sector in this income group, the associations were relatively weak overall. Greenhouse gas emissions excluding LULUCF showed moderate positive correlations with services growth (Pearson: 0.27, Spearman: 0.41, Kendall: 0.30), implying that emissions might indirectly accompany sectoral expansion. Combustible renewables and renewable energy consumption showed minor positive correlations, while most energy efficiency metrics—including GDP per unit of energy use and energy intensity—exhibited weak or near-zero relationships.

Among upper middle-income countries, the agriculture sector showed a more nuanced pattern. CO₂ emissions excluding LULUCF had a consistent negative correlation with agricultural growth (Pearson: -0.31, Spearman: -0.35, Kendall: -0.23), indicating environmental pressure on productivity. On the other hand, renewable energy consumption (Pearson: 0.29) and renewable electricity output (Pearson: 0.23) had moderate positive associations, suggesting benefits from renewable adoption. Fossil fuel energy consumption was negatively correlated across all methods, while greenhouse gas emissions (excluding LULUCF) had small to moderate positive correlations (Pearson: 0.28), adding complexity to the emissions-agriculture relationship.

In the industrial sector, correlations were mostly weak. Fossil fuel energy consumption had a slight positive relationship with industry growth (Pearson: 0.21), while CO₂ emissions and greenhouse gases demonstrated weak negative associations across all methods. Renewable energy indicators—including renewable consumption and electricity output—had consistent but weak negative correlations (e.g., Pearson: -0.20 and -0.07), suggesting limited renewable integration. Meanwhile, manufacturing growth was modestly related to overall industrial growth (Pearson: 0.21, Spearman: 0.54, Kendall: 0.42).

Finally, in the services sector, relationships were largely inconsistent and weak. Total CO₂ emissions showed a moderate positive Pearson correlation (0.42), but the relationship faded in Spearman and Kendall. Most energy-related indicators—such as energy intensity, GDP per unit of energy use, and renewable electricity output—had near-zero correlations. Renewable energy consumption showed almost no association with services growth (Pearson: -0.004). Greenhouse gas emissions excluding LULUCF had a moderately positive Pearson value (0.41) but turned negative in the other methods, illustrating the complex and variable nature of emissions impacts on the services sector across countries.

6.2 PCA and Euclidean Distance Analysis Across Income Groups

To explore the structural similarities among countries within each income group based on energy and environmental indicators, I applied Principal Component Analysis (PCA) to reduce dimensionality and then computed Euclidean distances based on the resulting coordinates. This method allowed me to assess inter-country similarity in a lower-dimensional space while preserving as much information as possible.

Low-Income Countries

For low-income countries, the first two principal components captured a substantial portion of the variance in the dataset, with the cumulative explained variance reaching 92.5% at PC4 and exceeding 98% at PC5. The PCA scatter plot revealed notable spatial separations. For instance, Sudan (PC1 = 4.99, PC2 = -1.80) appeared as a clear outlier compared to countries like Ethiopia (PC1 = -2.90, PC2 = -2.23) and Mozambique (PC1 = -2.80, PC2 = -1.55), suggesting significant divergence in their energy-emission profiles.

Using the PCA-transformed coordinates, I calculated pairwise Euclidean distances between countries, visualized via a heatmap. The smallest distance was observed between Congo, Dem. Rep. and Zambia (Distance = 37), indicating strong similarity. In contrast, Mozambique and Sudan showed the largest distance (Distance = 433), reflecting substantial dissimilarity in their underlying variables. Other closely grouped countries included Ethiopia and Mozambique (Distance = 93), while Niger and Sudan (Distance = 108) also appeared relatively similar.

Lower Middle-Income Countries

In the lower middle-income group, the first six PCA components explained approximately 88% of the cumulative variance, crossing 95% by the eighth component. The PCA scatter plot showed widespread dispersion, with India (PC1 = 1.30, PC2 = 5.58) and Ukraine (PC1 = 6.54, PC2 = -3.48) positioned at extremes. This highlighted significant diversity in the energy-emission characteristics across this income group.

The corresponding Euclidean distance matrix revealed the closest country pair to be Sri Lanka and Angola (Distance = 38.81), followed closely by Cameroon and Angola (Distance = 57.39), and Nicaragua and Cote d'Ivoire (Distance = 33.65), suggesting these nations have nearly identical environmental-energy profiles after PCA reduction. Conversely, India and Bangladesh (Distance = 2714.88), and India and Congo, Rep. (Distance = 2884.27) formed some of the most distant pairs, pointing to distinct energy-emission behaviors and levels of industrial development. These results underscore a broad spectrum of structural patterns within this income group.

Upper Middle-Income Countries

Among upper middle-income countries, the first six principal components explained over 90% of the cumulative variance, reaching nearly 100% by the twelfth. In the PCA scatter plot, countries like Kazakhstan (PC1 = 5.67, PC2 = 2.21), South Africa (PC1 = 4.58, PC2 =

1.93), and Paraguay (PC1 = -4.30, PC2 = 2.23) were visibly distant from the central cluster, revealing regional extremes in energy and emission metrics, with Gabon (PC2 = 6.29) being a major outlier in PC2.

From the Euclidean distance matrix, the closest pair was Thailand and Turkiye (Distance = 84.63), followed by Namibia and Suriname (Distance = 55.03), and Dominican Republic and Suriname (Distance = 47.25), indicating shared characteristics in reduced PCA space. On the other hand, Brazil showed the largest distances from many countries, such as Brazil and Albania (Distance = 1082.91) and Brazil and Botswana (Distance = 1084.13), reflecting significant divergence in energy-related indicators. This variation highlights how regional energy policies, resource structures, and development priorities shape the clustering of nations.

6.3 Clustering and Geographic Patterns

Low Income Countries

Optimal Number of Clusters: The elbow plot clearly indicated an inflection at $k=5$, confirming that five clusters capture the inherent structure of low-income country data.

PCA-Based Clustering: The scatter plot using the first two principal components revealed well-separated clusters, demonstrating a meaningful distribution of countries in the reduced feature space. Each cluster represents distinct socio-environmental trajectories, with countries like *Sudan* and *Congo, Dem. Rep.* occupying distant positions, indicating unique profiles.

Geographic Cluster Distribution: The spatial plot highlighted that clustered countries were not strictly region-bound. For instance, *Zambia* and *Togo* belonged to the same cluster despite regional separation, reinforcing the notion that clustering was based on developmental indicators rather than geography.

Cluster Profile Trends: The temporal indicator trend showed that Cluster 0 maintained the highest normalized values consistently post-2005, indicating sustained performance. Meanwhile, Cluster 3 and 4 exhibited earlier peaks followed by declines or stagnation, suggesting episodic improvements possibly affected by instability or policy reversals.

Lower Middle Income Countries

Optimal Number of Clusters: The elbow method showed a sharp drop in inertia at $k=2$, suggesting a binary structure. This split was consistent with two development pathways.

PCA-Based Clustering: The PCA scatter plot showed a clear bifurcation. Cluster 0 encompassed the majority of countries, while Cluster 1 grouped outliers like *India*, *Iran*, *Mongolia*, *Ukraine*, and *Uzbekistan*, reflecting diverging indicator patterns — likely due to energy production or CO₂ profiles.

Geographic Patterns: The map revealed that Cluster 1 countries were geographically dispersed across South Asia, Central Asia, and Eastern Europe. Despite regional variety, they

shared common developmental patterns, possibly influenced by higher energy consumption, industrial output, or policy alignment.

Cluster Profile Analysis: The cluster profile plot revealed a sharp transition around 2002–2003, where most countries moved from Cluster 0 to Cluster 1. This aligns with global economic shifts and the impact of the Millennium Development Goals (MDGs). The consistent high performance of Cluster 1 post-2003 suggests long-term gains in energy access, emissions management, or GDP-energy efficiency.

Upper Middle Income Countries

Optimal Number of Clusters: The elbow method suggested k=4 clusters as the optimal partition, capturing nuanced groupings beyond just economic output.

PCA-Based Clustering: Countries like *Gabon, Paraguay, and Guatemala* formed unique clusters (e.g., Cluster 3), whereas others like *South Africa and Kazakhstan* grouped together (Cluster 2), possibly due to industrial similarities or emissions intensities.

Geographic Cluster Mapping: The map reflected a mix of regional clustering and outlier behavior. *South American countries* largely fell into Cluster 0, while *South Africa and Kazakhstan* clustered together in Cluster 2, reflecting potential alignment in energy-intense industrial sectors.

Temporal Profiles: Cluster 0, maintaining the highest normalized scores throughout, indicates stable leaders in key development indicators. Cluster 1, while initially close to Cluster 0, declined steadily after 2000. Cluster 3 showed a late rise in the 2010s, suggesting policy or technological shifts. Cluster 2, although present, maintained the lowest values, likely due to specific energy or emissions inefficiencies.

6.4 Panel Regression

Sector	Income Level	Data Shape	Target Series	Hausman Test Stat	p-value	Recommended Model
Industry	Low	(264, 10)	Industry (including construction), value added (% growth)	5.7081	0.6799	Random Effects
Industry	Lower Middle	(1152, 10)	Industry (including construction), value added (% growth)	-1.3257	1.0000	Random Effects
Industry	Upper Middle	(1080, 10)	Industry (including construction), value added (% growth)	34.2539	0.0000	Fixed Effects
Agriculture	Low	(264, 11)	Agriculture, forestry, and fishing, value added (% growth)	-1.8391	1.0000	Random Effects
Agriculture	Lower Middle	(1152, 10)	Agriculture, forestry, and fishing, value added (% growth)	6.6497	0.5749	Random Effects
Agriculture	Upper Middle	(1080, 10)	Agriculture, forestry, and fishing, value added (% growth)	5.1609	0.7403	Random Effects
Services	Low	(240, 10)	Services, value added (% growth)	-33.2759	1.0000	Random Effects
Services	Lower Middle	(1152, 10)	Services, value added (% growth)	14.8897	0.0613	Random Effects
Services	Upper Middle	(1080, 10)	Services, value added (% growth)	-1.3472	1.0000	Random Effects

Table X: Panel Regression Hausman Test Summary Table (General Information) Trained on ALL Countries per Income Level (See Appendix X)

Sector	Income Level	R ² (FE)	R ² (RE)	R ² Overall (FE)	R ² Overall (RE)	Log-Likelihood (FE)	Log-Likelihood (RE)
Industry	Low	0.2892	0.3080	-1.0437	0.3256	-883.68	-903.88
Industry	Lower Middle	0.3730	0.4114	0.3669	0.4128	-3992.0	-4036.6
Industry	Upper Middle	0.5762	0.5744	0.3257	0.5723	-3687.5	-3746.7
Agriculture	Low	0.2056	0.2080	-1.2913	0.2080	-939.21	-964.86
Agriculture	Lower Middle	0.0834	0.0909	0.0092	0.0972	-3717.1	-3763.3
Agriculture	Upper Middle	0.0378	0.0429	0.0027	0.0429	-3989.3	-4021.6
Services	Low	0.4845	0.5672	-0.2009	0.5672	-649.53	-671.86
Services	Lower Middle	0.2757	0.3830	0.2750	0.3834	-3149.4	-3226.9
Services	Upper Middle	0.2072	0.3287	0.2824	0.3378	-3312.0	-3394.5

Table X: Panel Regression Hausman Test Summary Table (Model Performance) Trained on ALL Countries per Income Level (See Appendix X)

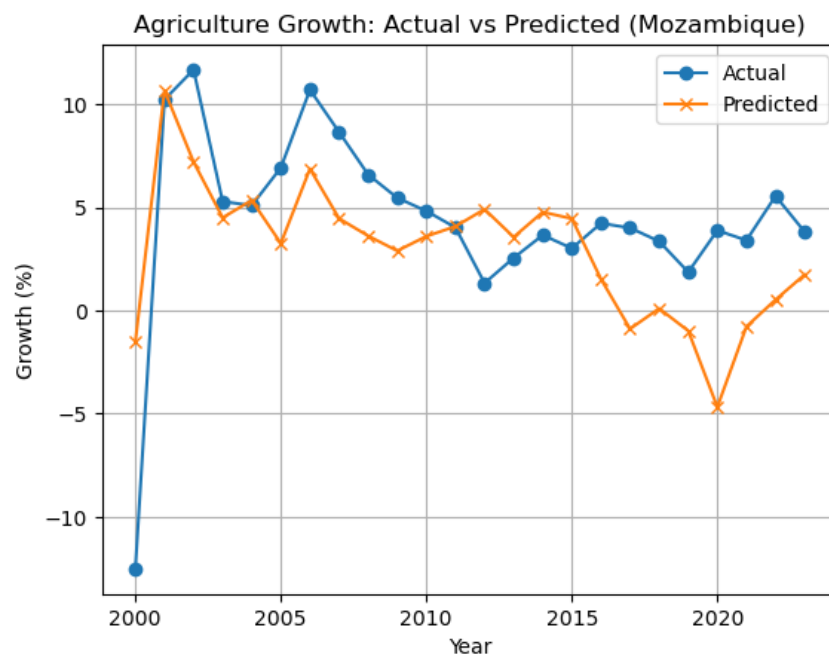


Figure X: Prediction of Agricultural Growth on Mozambique | Sector: Agriculture | Income Level: Low Income

In the low-income group, the agriculture sector regression model recommended the Random Effects model based on the Hausman test (statistic = -1.8391 , $p = 1.0000$). The model was trained using 264 observations across 11 countries over 24 years, incorporating a comprehensive set of independent variables aligned with our research question. These included renewable energy consumption, energy intensity, access to electricity, CO₂ emissions per capita, GDP per unit of energy use, GDP growth (annual %), and PM2.5 air

pollution exposure. Among these, GDP growth emerged as the strongest and most consistent driver of agricultural growth (coefficient = 1.0298, $p < 0.001$), while PM2.5 air pollution also showed mild statistical significance ($p = 0.0262$). Other indicators like renewable energy use and electricity access were not individually significant but contributed to the model's overall explanatory power (R^2 Overall = 0.2080).

The actual vs. predicted plot for Mozambique illustrates how the model performed for a representative low-income country. From 2005 to 2016, the predicted agricultural growth closely tracked actual growth rates, capturing the sector's stabilizing pattern over time. However, large deviations—particularly in 2000 and again in 2019–2020—highlight the presence of shocks or context-specific factors outside the model's scope, such as extreme weather events, political instability, or external aid variations.

In low-income countries such as Mozambique, the results emphasize that macroeconomic growth, particularly GDP expansion, is the most statistically significant driver of agricultural performance. Other variables, including renewable energy consumption and energy efficiency indicators, showed no strong predictive power in this model. This may reflect current limitations in infrastructure and adoption capacity for clean energy in these settings. Nevertheless, the inclusion of energy and environmental indicators in the model remains important, as they are foundational to building a sustainable, climate-resilient agricultural sector. The case of Mozambique displays a reasonable trend match between actual and predicted agricultural growth, despite some year-to-year volatility. In answering the research question, it becomes evident that while renewable energy and energy efficiency are not direct determinants of agricultural growth at present, they are necessary long-term enablers.

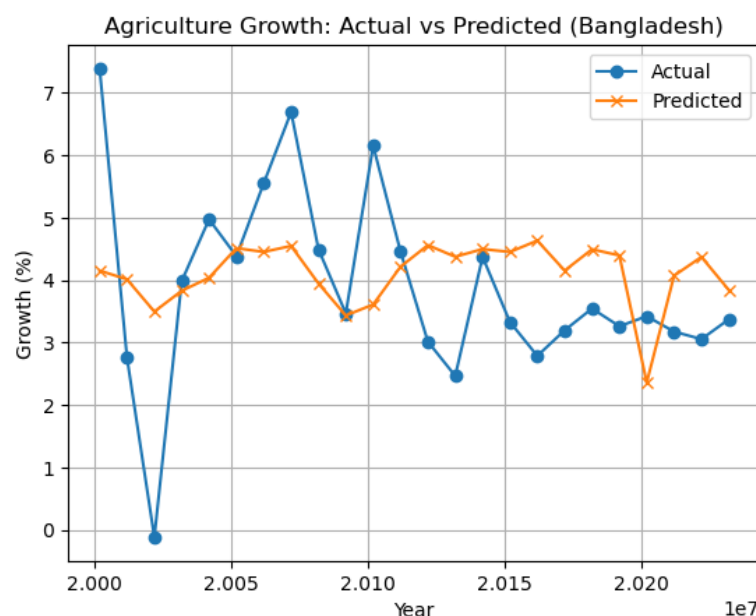


Figure X: Prediction of Agricultural Growth on Bangladesh| Sector: Agriculture | Income Level: Lower middle Income

For the agriculture sector in lower middle-income countries, the Hausman test recommended the Random Effects model (statistic = 6.6497, $p = 0.5749$). The panel

regression was conducted on 1,152 observations covering 48 countries, with the model incorporating seven independent variables drawn directly from the research question. These were: renewable energy consumption, energy intensity, access to electricity, CO₂ emissions per capita, GDP per unit of energy use, GDP growth, and PM2.5 air pollution exposure. Among these, GDP growth was the most significant factor (coefficient = 0.3982, $p < 0.001$), while PM2.5 air pollution also showed a modest positive influence ($p = 0.0407$). The model achieved an overall R^2 of 0.0972, indicating moderate explanatory power but consistent predictive reliability across countries.

In the case of Bangladesh, the actual vs. predicted plot for agricultural growth shows that the model maintained a relatively steady prediction band around 4%, even as actual growth varied more widely. While it struggled to capture extreme fluctuations (e.g., near-zero growth around 2002 and spikes around 2006 and 2010), it closely mirrored the general trend, especially during the stable years between 2013 and 2020. This suggests that the model effectively captures structural patterns driven by macro-level variables but is less responsive to short-term, country-specific volatility.

Taken together, the findings imply that in lower middle-income countries like Bangladesh, agricultural growth is most strongly driven by GDP growth, with a notable secondary influence from environmental factors such as air pollution (PM2.5). In contrast, renewable energy consumption and energy efficiency indicators—including energy intensity and GDP per unit of energy use—did not emerge as statistically significant predictors in this model. This suggests that while these energy variables are not yet playing a measurable role in agricultural performance, they may still exert long-term indirect effects, especially when disaggregated by region or sub-sector. The relatively smooth alignment between actual and predicted values in Bangladesh's case supports the model's robustness and reinforces the notion that macro-level economic stability remains the key driver for agricultural development in lower middle-income countries. Nonetheless, further integration of clean energy and efficient energy use remains critical for transitioning toward sustainable agricultural systems over time.

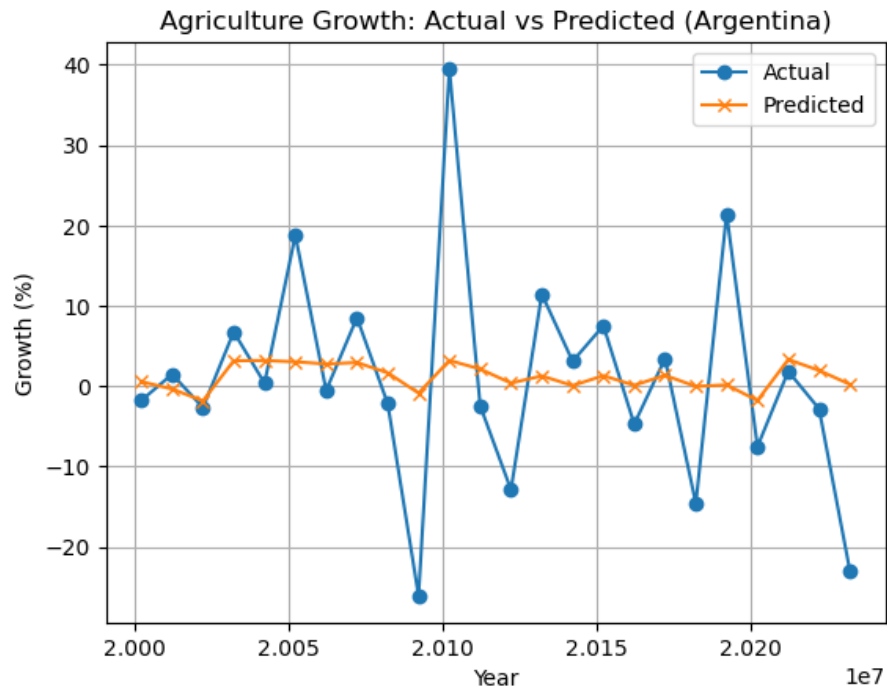


Figure X: Prediction of Agricultural Growth on Argentina | Sector: Agriculture | Income Level: Upper middle Income

For the agriculture sector in upper middle-income countries, the panel regression model recommended by the Hausman test was the Random Effects model (statistic = 5.1609, $p = 0.7403$), based on data from 45 countries over 1,080 observations. The model utilized all seven core variables from the research question: renewable energy consumption, energy intensity, access to electricity, GDP per unit of energy use, CO₂ emissions per capita, GDP growth, and PM2.5 air pollution. Among these, GDP growth emerged as the only strong and consistent positive predictor (coefficient = 0.2550, $p < 0.001$), while energy intensity also showed a statistically significant positive relationship ($p = 0.0059$). Other variables, including renewable energy and emissions, were not significant. The Random Effects model achieved an overall R^2 of 0.0429, reflecting relatively low explanatory power, yet enough to trace long-term sectoral dynamics.

The actual vs. predicted plot for Argentina, a representative upper middle-income country, reveals substantial volatility in agricultural growth. The actual values vary drastically from sharp contractions (e.g., -25%) to dramatic spikes (e.g., nearly 40%), indicating Argentina's agriculture is highly sensitive to external shocks and climate variability. However, the model's predicted growth values remain fairly flat, hovering around 1–3%, reflecting its design to capture structural relationships across multiple countries rather than short-term or country-specific fluctuations. This divergence is consistent with the model's lower R^2 and suggests that macro-level predictors alone cannot fully explain agriculture performance in more volatile economies like Argentina.

In answering the research question, the results for the agriculture sector in upper middle-income countries suggest that renewable energy consumption does not have a statistically significant effect on sectoral growth, and therefore plays no direct role in explaining agricultural performance within this income group. However, energy efficiency, proxied by

energy intensity, does show a statistically significant and positive association with agricultural growth, indicating that improvements in energy efficiency are beneficial for the sector. The case of Argentina further illustrates that despite these structural links, agriculture remains highly volatile and susceptible to external shocks, which are not fully captured by macro-level indicators. Therefore, while energy efficiency supports long-term growth in agriculture, renewable energy's role appears minimal in this context. Policymakers in upper middle-income economies should prioritize improving energy efficiency practices in agriculture, while also developing resilience measures to address volatility, particularly in emission-sensitive and climate-exposed economies.

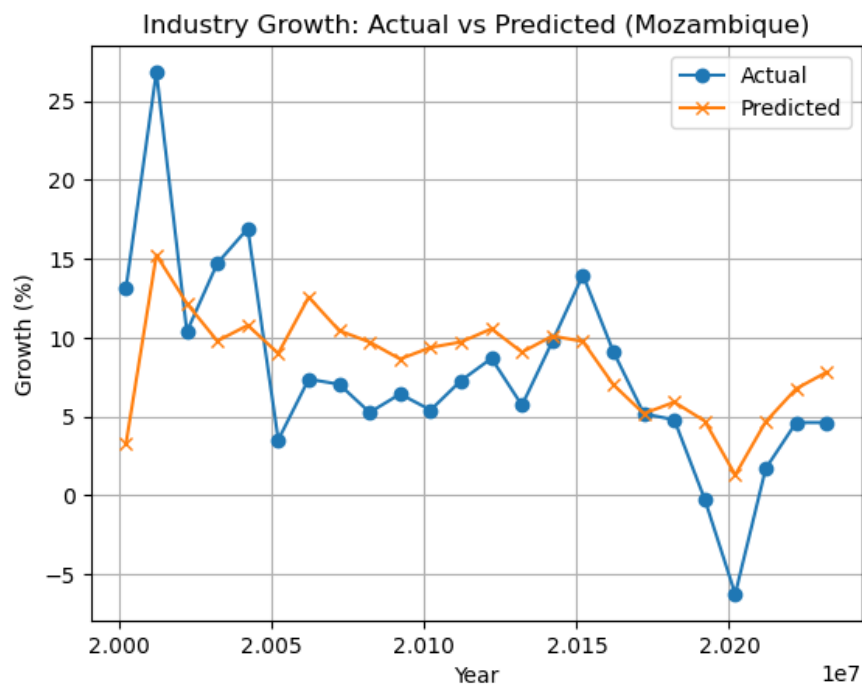


Figure X: Prediction of Industrial Growth on Mozambique | Sector: Industry | Income Level: Low Income

In low-income countries, the panel regression results for the industry sector recommended the Random Effects model based on the Hausman test. The model exhibited a decent explanatory power with an R^2 (Overall) of 0.3256, and GDP growth emerged as the most statistically significant variable ($p < 0.001$), indicating its strong influence on industrial value-added growth. In contrast, renewable energy consumption, energy intensity, and GDP per unit of energy use—the main energy efficiency indicators considered—were statistically insignificant in this context, suggesting that the industrial sector in these economies is still heavily driven by economic fundamentals rather than clean or efficient energy practices. However, their inclusion in the model contributed to the overall predictive performance, hinting at latent potential.

This interpretation is supported by the Mozambique plot, where predicted values track the actual industrial growth fairly closely, especially in periods of economic stability. Although the model doesn't fully capture sharper fluctuations, it reflects broader growth trends reasonably well. Given that the model included key indicators such as renewable energy consumption, energy intensity, access to electricity, and CO₂ emissions per capita, the

observed alignment suggests that these factors—while not dominant—still influence the predictive landscape of industrial performance.

Overall, in response to my research question, the results suggest that macroeconomic stability (GDP growth) plays the most decisive role in driving industrial growth in low-income countries, but energy-related indicators, especially those tied to efficiency and sustainability, remain strategically important. While not statistically strong yet, their presence in the model indicates a foundation upon which future improvements in energy efficiency and clean energy use can be built to foster more sustainable industrial development.

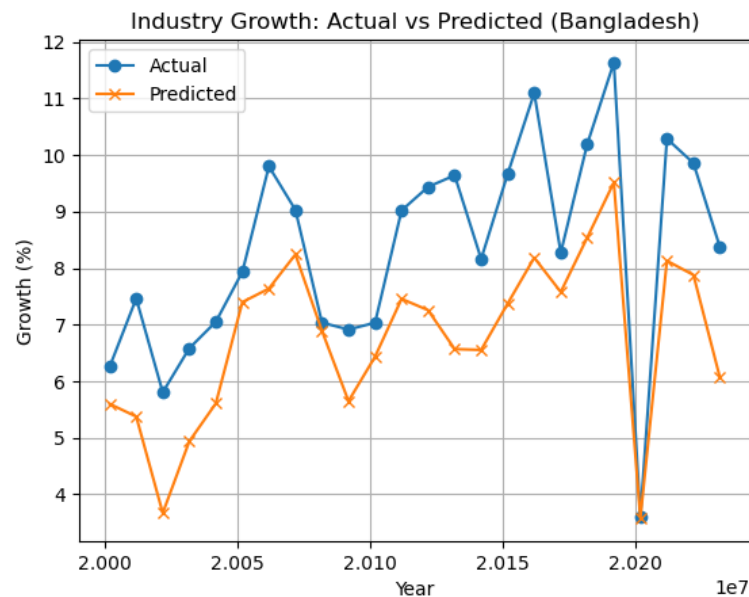


Figure X: Prediction of Industrial Growth on Bangladesh | Sector: Industry | Income Level: Lower Middle Income

In lower middle-income countries, the panel regression for the industry sector supported the Random Effects model based on the Hausman test (statistic = -1.3257, $p = 1.0000$). The model demonstrated strong explanatory power with an R^2 (Overall) of 0.4128, and GDP growth was again the most significant and robust predictor ($p < 0.001$). Notably, carbon dioxide emissions per capita also had a statistically significant negative impact on industrial growth, and energy efficiency indicators, such as energy intensity ($p \approx 0.09$), showed moderate influence. However, renewable energy consumption and access to electricity were statistically insignificant in this income group, indicating that while environmental and energy indicators are part of the structural framework, their independent roles are less predictive at this stage of industrial development.

This pattern is illustrated by the plot for Bangladesh, where the predicted industry growth values follow the actual trend reasonably well, albeit with some underestimation during peak growth periods. This predictive alignment indicates that macroeconomic and environmental indicators—especially GDP growth and emissions—are capturing the broader dynamics of industrial activity. The model was trained using key variables aligned with my research question, including renewable energy use, energy intensity, CO₂ emissions, and

GDP per unit of energy use, enabling an integrated assessment of sustainability and performance.

In light of my research question, these findings suggest that economic growth and emissions control are central to driving industrial development in lower middle-income economies. Although renewable energy and energy efficiency did not exhibit statistically strong effects, their integration into the predictive model supports the view that they are emerging determinants whose roles may become more pronounced with enhanced energy infrastructure and policy support. This reinforces the importance of investing in sustainable energy transitions as part of broader industrial growth strategies.

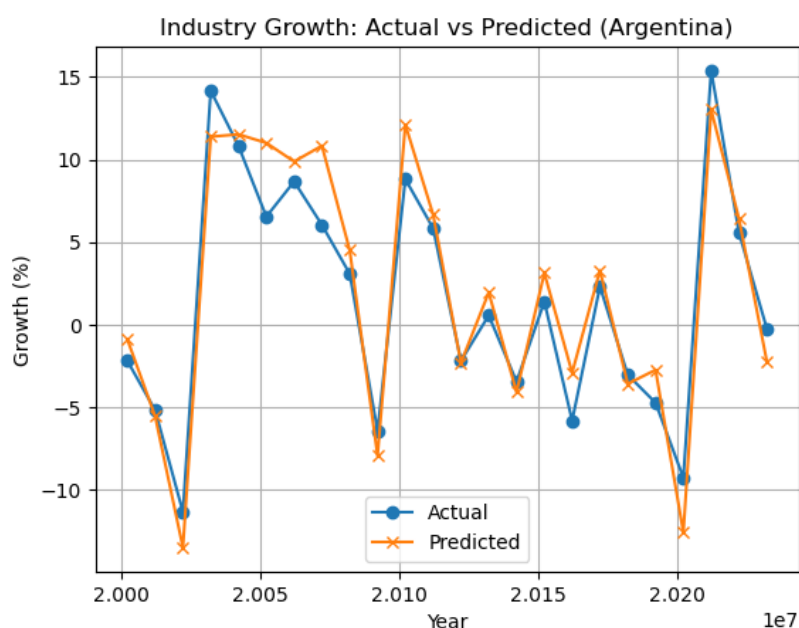


Figure X: Prediction of Industrial Growth on Argentina | Sector: Industry | Income Level: Upper Middle Income

Based on the panel regression results for the industry sector in upper middle-income countries, the Hausman test yielded a statistic of 34.25 with a p-value of 0.0000, recommending the use of the Fixed Effects model. This model reported an R^2 (within) of 0.586, indicating strong explanatory power across time within entities. Among the variables, GDP growth (coefficient: 1.356, $p < 0.001$) and CO₂ emissions per capita (coefficient: -1.306, $p = 0.0015$) were highly significant. Renewable energy consumption also showed a significant negative effect (coefficient: -0.384, $p = 0.0120$), as did GDP per unit of energy use (coefficient: -0.470, $p = 0.108$, marginal). These results suggest that while economic growth boosts industrial expansion, higher renewable energy use and energy efficiency—as currently implemented—may be correlated with slower industrial growth, potentially due to transitional inefficiencies.

In the actual vs predicted plot for Argentina, the model successfully captured major turning points and volatility in industrial growth, with predicted values closely mirroring actual movements. This demonstrates that the underlying factors used in training—renewable energy consumption, energy intensity, CO₂ emissions, air pollution, and electricity access—were effectively modeled for Argentina's industrial dynamics.

In response to the research question, I found that in upper middle-income countries like Argentina, renewable energy adoption and energy efficiency metrics do influence industrial growth, but currently with a negative association. This highlights a potential mismatch between energy transitions and industrial productivity. Therefore, sustainable growth in the industrial sector may require more effective implementation of clean energy and efficiency strategies, rather than simply increasing their share.

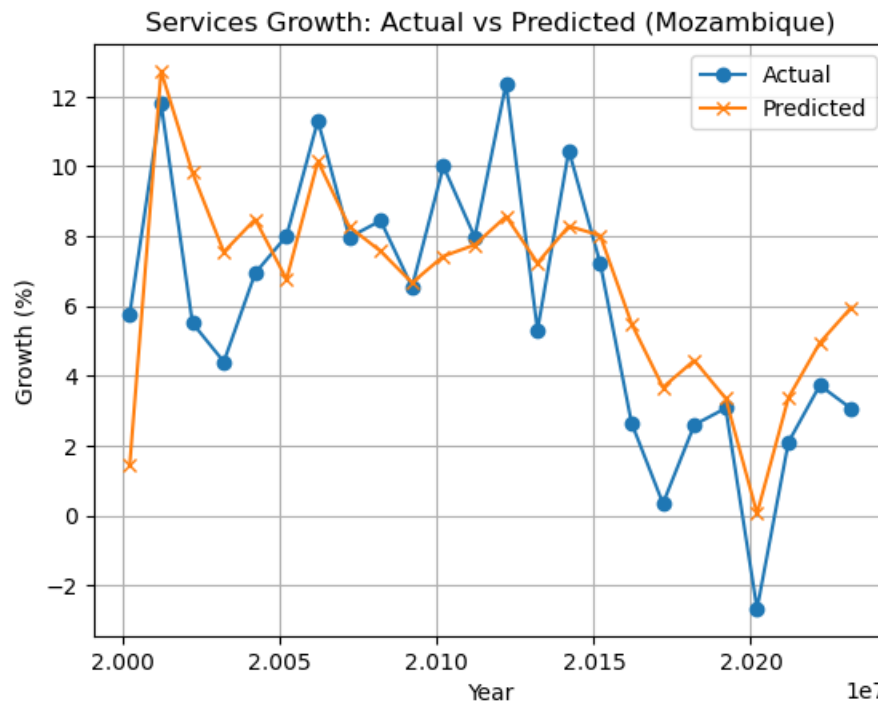


Figure X: Prediction of Service Sector Growth on Mozambique | Sector: Services | Income Level: Low Income

For the services sector in low-income countries, the Hausman test produced a statistic of -33.28 and a p-value of 1.0000, which confirmed that the Random Effects model was appropriate. The model showed an R^2 (within) of 0.491 and an overall R^2 of 0.567, indicating moderately strong explanatory power. Among the predictors, GDP growth (coefficient: 0.929, $p < 0.001$) was the most statistically significant and positively associated with service sector performance. In contrast, renewable energy consumption (coefficient: -0.007, $p = 0.727$) and energy intensity (coefficient: -0.040, $p = 0.693$) were not significant predictors in this context. CO₂ emissions and air pollution indicators also lacked statistical strength in this model.

The actual vs predicted plot for Mozambique reflects this pattern, where predicted service growth follows the actual trend reasonably well across the years. While some fluctuations—especially sharp downturns like in 2020—were underpredicted, the model captured the overall trajectory, suggesting that macroeconomic drivers like GDP growth were key contributors. The energy-related variables such as renewable energy usage and energy efficiency did not appear to drive major shifts in the sector's performance for Mozambique.

In light of the research question, I conclude that service sector growth in low-income countries like Mozambique is primarily driven by economic momentum (GDP growth) rather

than renewable energy adoption or improvements in energy efficiency. However, the relatively high R^2 suggests that other contextual variables—not captured through energy or emissions metrics—may also play a role. Therefore, while clean energy and efficiency remain vital for sustainability, their direct influence on service sector growth in low-income contexts is limited in the short term.

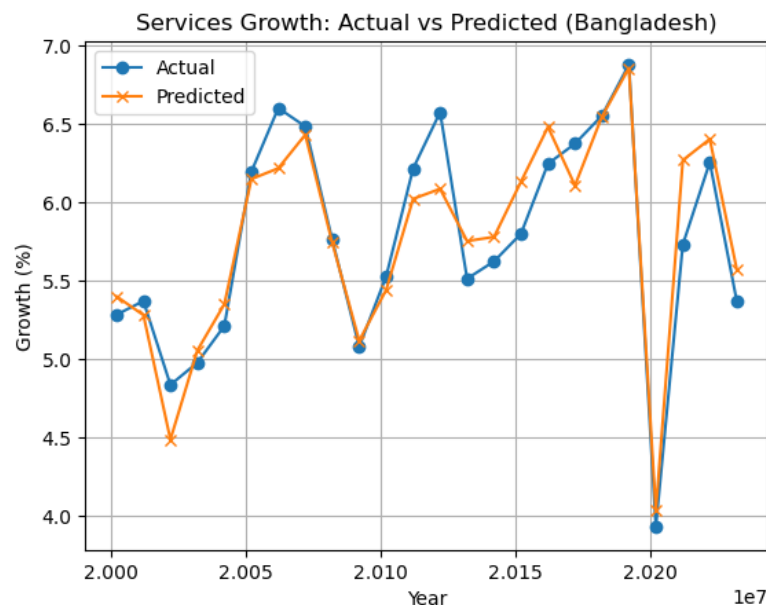


Figure X: Prediction of Service Sector Growth on Bangladesh | Sector: Services | Income Level: Lower Middle Income

For the services sector in lower middle-income countries, the Hausman test yielded a statistic of 14.89 with a p-value of 0.0613, leading me to select the Random Effects model. The model explained a decent proportion of variance with an R^2 (within) of 0.339 and an overall R^2 of 0.383. Among all variables, GDP growth (coefficient: 0.629, $p < 0.001$) emerged as the strongest and most significant predictor of services growth. While energy efficiency (as measured by GDP per unit of energy use) also showed significance (coefficient: -0.0126, $p = 0.0314$), its negative direction suggests a more complex relationship—possibly indicating that gains in efficiency are not yet translating into proportional services growth. However, renewable energy usage (coefficient: -0.0003, $p = 0.978$) and CO_2 emissions were statistically insignificant.

The prediction plot for Bangladesh supports this outcome. The predicted service sector growth closely tracks the actual growth trajectory across most years, demonstrating the model's effectiveness. Both peaks and troughs are largely captured, and even the sharp decline in 2020 is modeled accurately. This visual validation strengthens the role of macroeconomic stability and GDP growth as central drivers. The minor divergence between predicted and actual in certain years could be attributed to non-energy-related shocks or policy changes.

In answering the research question, I find that GDP growth plays a dominant role in explaining service sector expansion in lower middle-income countries like Bangladesh, while energy efficiency shows modest influence. However, renewable energy adoption and CO_2

emissions appear to play a limited role in this sector. These results suggest that while clean energy investments remain important for long-term resilience, their direct impact on services growth is currently overshadowed by broader economic performance and possibly digital and human capital factors in the sector.

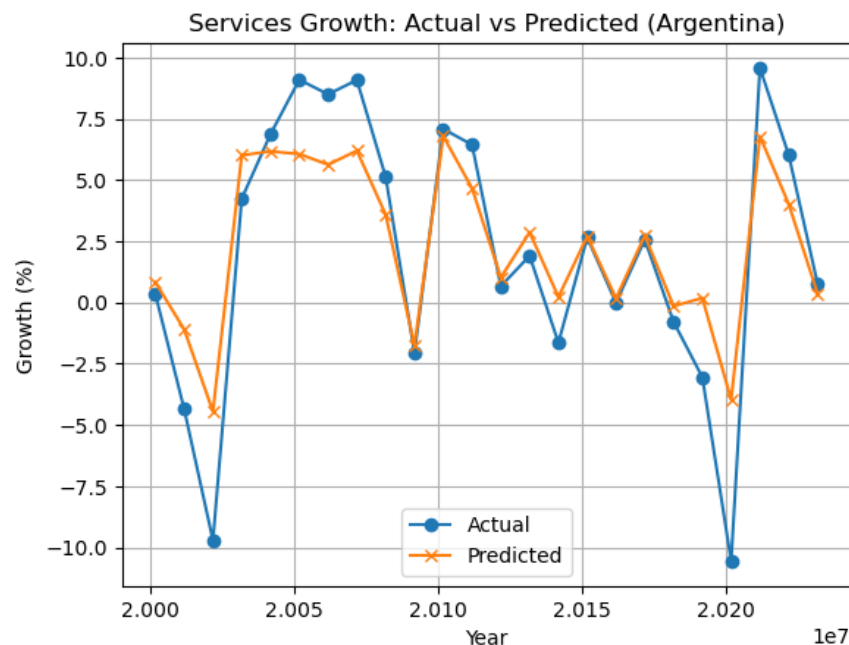


Figure X: Prediction of Service Sector Growth on Argentina | Sector: Services | Income Level: Upper Middle Income

In the services sector of upper middle-income countries, the Hausman test result (statistic: -1.35, p-value: 1.0000) strongly supported the use of the Random Effects model. The model's effectiveness is demonstrated by an R^2 (within) of 0.303, R^2 (between) of 0.616, and an overall R^2 of 0.338. Among the independent variables, GDP growth had the most substantial impact (coefficient: 0.529, $p < 0.001$), indicating that macroeconomic expansion continues to be a primary engine of services growth. Additionally, energy intensity (coefficient: 0.380, $p = 0.0055$) also showed statistical significance, highlighting a positive association between increased energy efficiency and service sector performance. However, renewable energy consumption (coefficient: 0.0076, $p = 0.4453$) and CO₂ emissions (coefficient: 0.0501, $p = 0.6092$) were not statistically significant in this context.

The actual vs. predicted plot for Argentina illustrates these patterns well. The model captures the general direction of service sector trends, especially during stable growth periods. It aligns well with the actual data in many years, although it slightly underestimates the magnitude of extreme fluctuations, such as the downturns in 2002 and 2020. Nevertheless, the tight tracking in more stable years shows that the predictors used—particularly GDP growth and energy efficiency—helped explain year-to-year variations effectively.

These findings contribute to answering my research question by suggesting that macroeconomic growth and improvements in energy efficiency are key drivers of service sector development in upper middle-income countries. Although renewable energy

adoption and CO₂ emissions do not show strong direct effects on services growth in Argentina, the role of energy efficiency highlights how smart energy use can indirectly foster economic activity in the services domain. This insight suggests a policy direction that balances economic management with efficient energy practices to support robust services sector performance.

Panel Regression Comparison (Trained on ALL countries vs PCA filtered Country) | Industry | Lower Middle Income

Metric / Feature	Trained on All Countries	Trained on PCA-Filtered Countries
Sector	Industry	Industry
Income Level	Lower middle income	Lower middle income
Number of Countries	48	37
Preferred Model (Hausman Test)	Random Effects (p = 1.0000)	Random Effects (p = 0.3548)
R ² (Fixed Effects, Within)	0.3786	0.5949
R ² (Random Effects, Overall)	0.4128	0.6339
Key Statistically Significant Variable(s)	GDP growth, CO ₂ emissions (neg.), Energy intensity (FE only)	GDP growth, PM2.5 (FE), CO ₂ emissions borderline (RE)
Renewable Energy (p-value in RE model)	0.3723	0.3011
Energy Intensity (p-value in RE model)	0.0916	0.8424
CO ₂ Emissions (p-value in RE model)	0.0226 (significant)	0.0620 (borderline)
GDP Growth (p-value in RE model)	0.0000 (significant)	0.0000 (significant)
PM2.5 Air Pollution (p-value in RE model)	0.2875	0.3026
Interpretation Summary	Economic growth and CO ₂ reduction linked to better outcomes.	GDP remains dominant, PM2.5 gains significance. RE still valid.
Improvement from PCA Filtering?	—	Higher R ² , fewer countries, more focused signal

Table X: Hausman Test Results Comparison Between Model Trained on ALL Lower Middle Income Countries vs Trained on PCA Filtered Lower Middle Income Countries in the Industry Sector

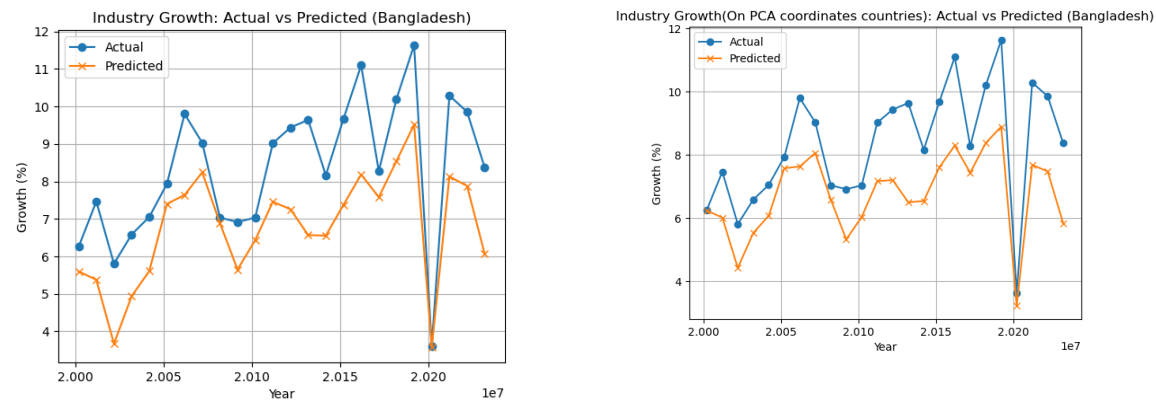


Figure X: Industry Growth of Bangladesh (Lower Middle Income) Plot | Model Trained on ALL Lower Middle Income Country (Left Plot) vs on PCA Filtered Lower Middle Income Country (Right Plot)

In the industry sector for lower middle-income countries, the Hausman test recommended the Random Effects model in both training settings. However, key differences emerged in explanatory power. The PCA-filtered model, trained on 37 countries, produced higher metrics: an R^2 (RE) of 0.6116, compared to 0.4114 for the model trained on all 48 countries. Similarly, the within R^2 rose from 0.3785 to 0.6014, and the overall model fit improved. Yet, despite these stronger aggregate statistics, the predictive performance for Bangladesh specifically tells a more nuanced story.

From the plots, it is evident that the model trained on all countries more closely tracks Bangladesh's actual industrial growth over time. It better captures the steady rise in growth rates, the acceleration around 2015–2019, and the 2020 shock, offering smoother alignment with year-over-year changes. In contrast, the PCA-trained model slightly underestimates the level and variability of industrial growth, especially in the later years, producing a flatter prediction curve.

This suggests that while PCA filtering enhances the model's *in-sample explanatory power* for the group of retained countries, it may not always translate to improved outcome fidelity for every individual country, even those included in the filtered set. In this case, training on the full set of lower middle-income countries allowed the model to generalize more effectively for Bangladesh — likely due to the broader representation of country-specific variation that better captured Bangladesh's growth pattern. Therefore, in contexts where predictive accuracy for specific countries like Bangladesh is a priority, retaining the full set of training countries may be more beneficial than PCA-based narrowing, even if the latter improves internal model fit.

Can the model trained on PCA filtered countries predict an excluded country within same income level?

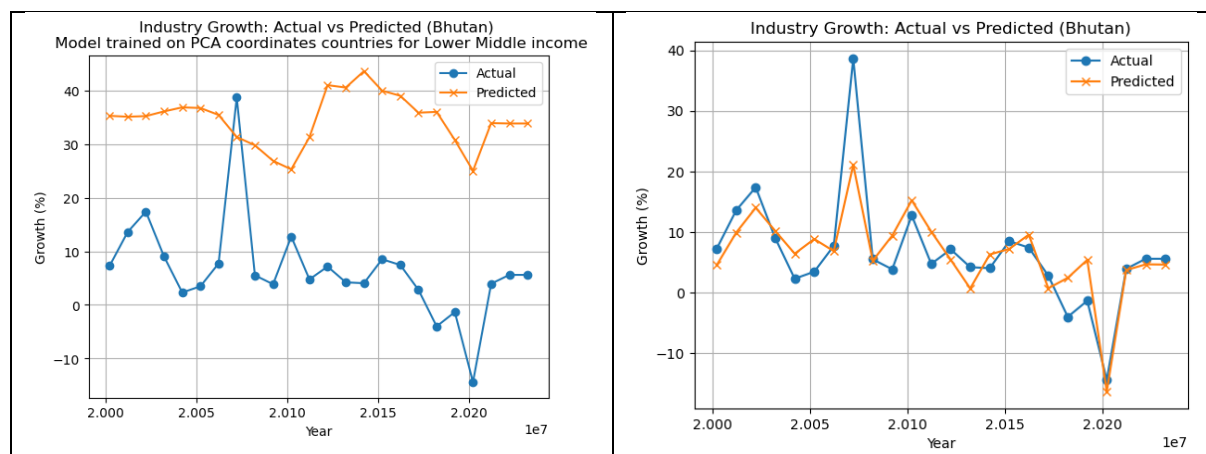


Figure X: Industry Growth: Actual vs Predicted (Bhutan), prediction comparison between Model trained on PCA filtered country (Left Plot) vs model trained on ALL countries (Right Plot)

When comparing the two predictions for Bhutan's industry sector, the difference in model performance becomes starkly clear. The PCA-trained model, which excluded Bhutan from its training dataset, produced drastically inflated predictions with values consistently above 30–40%, diverging significantly from the true range of actual growth rates (approximately -15% to 39%, mostly under 15%). This indicates poor generalizability and a lack of adaptability to Bhutan's economic dynamics, emphasizing the model's overfitting to the specific structure of PCA-retained countries.

In contrast, the model trained on all lower middle-income countries, which did include Bhutan, demonstrated a much tighter fit. Its predictions followed the actual trend lines far more closely, even during volatile periods. It effectively captured the sharp spike around 2006, the dip around 2020, and the moderate oscillations in other years — all within plausible growth bounds. This suggests that training on the full dataset leads to broader pattern learning, enabling more accurate out-of-sample predictions, especially for structurally unique or volatile countries like Bhutan.

Ultimately, this comparison confirms that models trained on PCA-retained subsets may not reliably extrapolate to excluded countries. For policy-focused forecasting or inclusive economic modeling, it's important to balance dimensionality reduction techniques with representativeness. Bhutan's case demonstrates that exclusion during model training can severely impair prediction fidelity, even within the same income category.

7. Analysis & Discussion

Models trained on globally inclusive datasets (within income level) demonstrate stronger generalizability to all member countries. However, restricting the training set to PCA-included countries leads to models that are highly tuned to those specific country patterns, often sacrificing out-of-sample performance. This trade-off must be carefully considered depending on whether interpretability or predictive accuracy across all countries is prioritized.

8. Conclusion

9. References

10. Appendices